



SYMBIOSIS

SCHOOL OF ECONOMICS

STIGLER'S NUTRITION PROBLEM (DIET)

Topic: This model determines a Least-Cost Diet which meets with Required Daily Allowance of nutrients for a moderately active adult woman weighing 55kgs.

Contributors:

Parsh Khandelwal - 25060242098

Riddhi Chowdhuri - 25060242109

Rupanjana Dey - 25060242118

Sneha Kulal - 25060242137

Sura Anjana Srimayi - 25060242145

Suvankar Das - 25060242146

Taniya Bamnodkar - 25060242151

Turnitin Report

Riddhi Chowdhuri - 25060242109 Report.pdf

ORIGINALITY REPORT

2%

SIMILARITY INDEX

1%

INTERNET SOURCES

1%

PUBLICATIONS

0%

STUDENT PAPERS

PRIMARY SOURCES

1

repositorio.ucv.edu.pe

Internet Source

<1%

2

Jianguo Liang, Akinobu Murata. "Analysis of the CCS Considering Environment Co-benefit of Air Pollutants in China", Energy Procedia, 2013

Publication

<1%

3

www.istat.it

Internet Source

<1%

4

Fakhimi, Ramin. "Advances in Discrete Optimization: From Truss Design Optimization to Quantum Computing Optimization", Lehigh University, 2024

Publication

<1%

5

jfrm.ru

Internet Source

<1%

6

niua.in

Internet Source

<1%

Exclude quotes On

Exclude matches < 14 words

Exclude bibliography On

INDEX

Abstract	3
Key words	3
Introduction	4
Objectives	6
Literature Review	7
Literature Gaps and future directions	9
Theoretical Framework	10
Policy Implications & Output Recommendations	19
Conclusion	21
References	22
Appendix	23

Abstract

This research uses the Classical Stigler Diet Problem, originally developed by George Stigler in 1945, to come up with a least-cost, nutritionally adequate diet, and customized to the dietary requirements of an Indian woman with a weight of 55 kilograms. The aim of the research is to determine the least cost-effective mix of popular food items that are consumed in India that will comply with the Recommended Daily Allowances (RDA) of the main nutrients, which include protein, fat, carbohydrates, vitamins, and minerals, at the minimum overall cost of the diet. In order to do it, a representative food basket consisting of cereals, pulses, milk, eggs, wheat, vegetable oils, nuts, and other appropriate foods that are characteristic of the Indian eating style are chosen. The formulated problem is a linear programming (LP) model, which is developed in the General Algebraic Modeling System (GAMS). In the model, nutrient requirements are constraints to make sure that the RDA of each nutrient is achieved with the objective function trying to minimize the overall price of the diet. Data of food prices and composition of nutrients are introduced to model realistic diet planning. The findings identify the viable opportunities of optimization methods in enhancing food policy and nutrition planning. The model offers information on designing balanced diets effectively, making them affordable and yet nutritionally sufficient. These results have wide implications on household budgeting, institutional meal planning in institutions like schools and hospitals, and national nutrition policies to help reduce malnutrition among women in India. This study by incorporating economic optimization and nutritional science highlights the importance of data-driven policies in policy making and resources allocation. All in all, the study shows that the classical framework suggested by Stigler with the design tailored to the modern Indian setting can be an effective instrument to create cost-efficient, nutritionally optimal diets that would not only contribute to personal health but also to national food security.

Key words

Stigler's Diet Problem, Indian Women Nutrition, Linear Programming, Gams model, Recommended Dietary allowances (RDA), Least Cost Diet, Women Health Economics.

Introduction

Having a proper diet is one of the pillars of health, productivity, and well-being. In developing nations like India, nutritional shortages, especially, micronutrient deficiencies, and undernutrition remain one of the key issues of public health (Rao, Balkrishna, Laxmaiah, & Venkaiah, 2015). The challenge of consumption of a balanced diet that meets the Recommended Dietary Allowances (RDA) at a cost low enough to be affordable to both the household and the policymaker is a consistent challenge to the households and policymakers. An increase in food prices, changes in diet, and socio-economic disparity further challenge the quest to achieve equitable food security and optimum nutritional results in dissimilar population groups.

The Diet Problem was originally created by George Stigler (1945) to determine the least-cost combination of foods that would meet minimum nutritional requirements. His innovative research was one of the first efforts of combining economics and human nutrition. George Dantzig subsequently re-stated the problem of Stigler in terms of Linear Programming (LP) methods, thus, becoming one of the classic models of optimization in economics and operations research (Dantzig, 1990). The Stigler Diet Problem has found extensive application since that time in various disciplines such as agricultural economics, health planning, the design of food policy, and resource allocation (Taha, 2017; Srinivas and Shastri, 2013). Dietary guidelines in the Indian scenario are mostly based on the Indian Council of Medical Research -National Institute of Nutrition (ICMR-NIN, 2020), where nutrient requirements are regularly revised according to age, gender, body weight and activity level. The RDA would prescribe about 1900 kilocalories of energy, 46 grams of protein and the amounts of important micronutrients like calcium, iron, iodine, vitamin C, vitamin D, niacin and vitamin B12 in a moderately active adult Indian woman of 55 kg body mass. These are some of the nutrition standards that constitute the constraint in the optimization model that will be applied in this study. The present study uses a classical framework proposed by Stigler to establish the application of mathematical optimization to inform the development of cost-effective diets in limited resource contexts, specifically applied to women in India. The model is developed through the Linear Programming in GAMS, whereby the minimum cost of the food mix of popular foods in India, including cereals, pulses, milk, eggs, oils, and nuts are identified to ensure the necessary nutrient levels are achieved. The results

are relevant to household budgeting, menu planning in institutions, and government nutrition policies, in particular, the policies that need to be designed to enhance the health of women and prevent diet-related disparities. Finally, the research is a linkage between economic modeling and nutritional science, which offers a data-driven methodology to construct affordable and nutritionally balanced diets in the Indian setting.

Objectives

1. To determine the least-cost diet for a 55 kg moderately active Adult Indian woman using linear programming.
2. To formulate and solve the diet problem in GAMS with constraints based on Indian RDA standards.
3. To compare nutrient adequacy and costs across food combinations.
4. To illustrate how optimization can inform food planning and public health policy for women.

Literature Review

The diet issue, reducing the cash outlay and satisfying the nutritional target, was first introduced by the seminal work of Stigler on the cost of subsistence that gave formal expression to the concept of finding a minimum cash diet that met a nutrient constraint set (Stigler, 1945). The empirical demonstration by Stigler had two key results: linear programming (LP) is a natural mathematical model of the problem, and unconstrained minima tend to give implausible, corner-solution diets (e.g., a small number of foods in large amounts) that though nutritionally complete are not practically feasible. The research by Stigler therefore forms the historical and theoretical foundation of modern research on LP diets.

The original development of linear programming by Dantzig provides both the formal means to formulate and solve the diet problem, and the interpretation of solver diagnostics (objective value, reduced costs, dual/shadow prices) in economic and nutritional terms (Dantzig, 1963). The presentation of duality and sensitivity analysis offered by Dantzig is one of the most important to diet modelling since it describes the process of determining which nutrient constraints are cost-driving and the sensitivity of results on demand or price changes that lead to the optimal solution. Applying the principles of Dantzig responsibly will help in transparency in reporting as well as in building model regularizations (e.g. variety constraints, upper bounds).

Garille and Gass (2001) re-examine the original calculations of Stigler with the current nutrient and price data and, what is more important, comment on the model extensions that generate more realistic diets. Their article shows (a) that revised food composition and price inputs can alter optimal diets, and (b) that the realistic solutions of pure LP models which include variety, minimum serving sizes, limits on the amount of one food a person should eat, integer constraints on discrete items, and formulations that are multi-objective and trade cost against palatability. This research directly influences real-world modelling decisions in a 55-kg woman: constraints that represent acceptable servings and food variety in that culture will eliminate the single-food-item diets that Stigler experienced.

In the case of food composition inputs in India, the most applicable, nationally representative database of nutrients is the Indian Food Composition Tables (IFCT 2017) (ICMR 2017). IFCT

contains nutrient profiles of hundreds of common foods eaten in India, and is better than non-local databases in modelling Indian diets. Since the nutrient is reported in edible weight (usually per 100 g) of an item, the practitioners should always follow a consistent unit transformation between a nutrient vector and a price data (e.g., rupees per 100 g) when building up the LP coefficient matrix. The range of IFCT allows culturally appropriate lists of candidate foods to be given to a 55-kg Indian woman so that the model can be representative of local foods and local food preparation practices.

Diet problems can be optimized to health goals (e.g. DASH diet and hypertension) and local settings based on applied LP diet research (e.g. Iwuji, Nnanna, and Ndulue, 2016). These applied papers usually use domain-specific nutrient limits, have upper/lower limits to make foods palatable and safe (e.g., sodium upper limits), and demonstrate the use of local price data is significant. They also show implementation options (defining decision variables in grams, servings, or per-rupee) which have a material impact on the interpretability and feasibility of a target person (say a 55-kg woman). Applied studies thus offer viable models on how to integrate NIN/IFCT nutrient standards with price variables and palatability limitations.

Synthesis and implications to modelling a 55-kg woman. Three lessons that can be taken out of the literature to develop an applied Stigler-style model are converged. To establish lower/upper nutrient limits, first use an authoritative nutrient standard which is relevant to your population (NIN RDAs or the Dietary Guidelines). Second, consider IFCT (2017) as the main food composition source of Indian foods and the USDA FDC as the supplement to the main source when needed. Third, avoid unrealistic corner solutions, introducing palatability/variety constraints (minimum number of different foods, realistic upper bounds, discrete servings of whole items), and solve LP duals to find costly nutrients. A combination of these practices gives a model which is computationally feasible, nutritionally justifiable, and culturally plausible in the case of a 55-kg woman.

Literature Gaps and future directions

The existing literature has quite a number of gaps. First, the majority of linear programming (LP) diet-related research is mainly single objective, aiming to minimize monetary cost, and few of them combine more encompassing objectives like eco-friendliness, cultural desirability, or limits on food preparation duration. This will limit the applicability of the diet problems in real world scenarios. Second, few LP-based diet formulations are parameterised to women of particular body mass - e.g. a 55 kg adult woman - based on individualised estimates of energy expenditure and gender-specific nutrient requirements. This disparity is especially prominent in the Indian context, since the nutrient needs and dietary patterns are very different, compared to the Western ones. Although the intended Indian women have been considered using nationally appropriate Recommended Dietary Allowances (RDAs) in the Indian Council of Medical Research-National Institute of Nutrition (ICMR-NIN), very few studies on optimization have directly employed the values in LP models. Finally, better interaction of the variability of food composition data across regions, as available in the Indian Food Composition Tables (IFCT 2017) and sophisticated LP methods like stochastic or interval programming is also needed. Adding these types of uncertainty modelling would enable researchers to consider natural differences in nutrient content between regions and cooking techniques. Sealing such gaps would make LP-based dietary models more culturally, nutritionally and policy relevant and produce more realistic, person-centred diet prescriptions to Indian women.

Theoretical Framework

The study answers one of the key questions of nutritional planning: What is the lowest-cost mix of foods that are readily available that can meet the daily nutritional needs of a moderately active 55 kg woman? A Linear Programming (LP) framework is applied to the problem, which allows making a dietary plan that is cost-effective and systematically determined on the basis of nutritional constraints.

When it comes to the issue of dietary design or the optimal diet as far as the field of public health and dietetics is concerned, the price of food is frequently a constraining factor. Linear programming model is a scientific approach to reducing the daily food spending, at the same time maintaining the necessary amount of essential nutrients in a diet, as the national and international health organizations suggest. These models have been extensively applied in the development of diets at the individual, hospital, school, and humanitarian levels, and make them an effective optimization tool to nutritional modeling within real world constraints.

The aim of the research is to come up with a mathematical model that minimizes the overall daily cost of what one consumes to sum up to the nutrient requirements. The model employs the use of the locally available food items and nutrient information to emulate the actual consumption trends. An LP is used to generate the daily cost-minimized diet, which is ensured to contain and surpass the Recommended Dietary Allowance (RDA) of intake of macronutrients (calories and protein) and micronutrients (iron, calcium, and vitamins) as required by adult women.

Linear Programming is a mathematical method of optimization of a linear objective (minimization) with equality or inequality constraints that are linear and non-negativity constraints. Problems related to the allocation of resources and cost optimization are often applied to the field of economics, engineering, logistics, and public health using LP. Overall, a Linear Programming Problem (LPP) has the following elements:

- Objective Function: Goal of the problem is a linear expression of the objective of the problem normally to maximize or minimize a particular value. This role in the nutrition problem aims at reducing the overall cost of daily diet.
- Decision Variables: The unknowns that the model attempts to solve. The decision variables in this case will be the sum of money (in rupees) to be spent per day on each food item.
- Constraints: They are linear equalities or inequalities that the decision variables should fulfill. They are in the form of nutritional needs, and the chosen type of food balance is to have the minimum amount of the most essential nutrients that are not below the RDA.
- Non-negativity Constraints: Decision variables are only allowed to be positive or zero since negative consumption or expenditure will not make sense in this scenario.

In the diet optimization model, the foods are taken as the pool of resources and each type of food adds a certain amount of each nutrient per unit of spending. Nutrient constraints are built on a nutrient composition table where the quantity of each nutrient content in 100 rupees of each food is determined. The LP model can be used to tell the optimal (least-cost) diet plan to use based on the nutritional needs. This formulation will not only make it possible to plan individualized dietary planning but will also provide meaningful information to policy-makers, healthcare professionals, and nutritionists in the creation of cost-effective nutrition planning at both individual and population levels.

Part I - Inputs

Sets: Two primary sets ‘Food’ (f) and ‘Nutrients’ (n) have been included in the model, altered to suit the Indian dietary preferences and patterns. The set ‘n’ lists the nutrients that must be consumed on a daily basis and hence acts as a constraint in the model. Set ‘f’ on the other hand contains the basket of food items typically consumed by Indians, that are the source of the required nutrients.

- Foods (f) : Milk, orange, apple, meat, leafy vegetable (spinach), eggs, millet, ghee, butter, vegetable oil, pulses, almonds, oilseeds, wheat and rice
- Nutrients (n) : Calories, calcium, protein, iron, iodine, vitamin C, niacin, vitamin B12.

Parameter b(n) : The required daily allowance (RDA) for the nutrients for an adult female, weighing 55 kg stated in set ‘n’ serves as the parameter of the model and can be considered as the lower bound in the constraint. This data has been sourced from (Dietary guidelines for NIN).

The RDA is as follows:

Calorie	1660kcal
Calcium	1000mg
Protein	46g
Iron	29mg
Iodine	140mg
Vitamin C	65mg
Niacin	11mg
Vitamin B12	2.2mg

Nutrient value for every 100 Rs a(f,n)

This is a two dimensional parameter (table) which specifies the amount of nutrient (n) provided by Rs.100 of food (f). The columns list the nutrients while the rows contain the food. The values have been derived as follows:

1. Obtained data for the nutrient composition for the foods in 'f'.
2. Then based on the average retail prices of foods, we calculated the amount of food that can be bought with Rs.100. (g) The formula used was:

$$(100/\text{Price of good}) \times 1000$$

3. To obtain the values in a(f,n) we calculated the amount of nutrients for every Rs.100 spent. The following formula was used:

$$(n/100) \times (g)$$

Where, $n/100$ = nutrient per gram of food; g = amount of food worth Rs.100

Food	Nutrients (per 100 Rs)							
	Calorie Kcal	Calcium mg	Protein g	Ironmg	Iodine ug	Vitamin C mg	Niacin mg	Vitamin B-12 ug
Milk	1140.350 877	2174.96	61.39	0.3508	420.96	35.08	3.508	7.016
Apple	399.88	46.14	1.9994	0.9228	0	35.374	0	0
Meat	872	28	64	4.8	0	0	54	0.0104
Green Leafy Vegetable	1000.04	4304.52	126.09 2	117.396	0	1221.788	0	0
Eggs	1347.06	527.52	118.69 2	16.956	0.02355	0	0	0.013188
Orange	587.5	500	11.75	1.25	0	665	0	0

Food	Nutrients (per 100 Rs)							
	Calorie Kcal	Calcium mg	Protein g	Ironmg	Iodine ug	Vitamin C mg	Niacin mg	Vitamin B-12 ug
Milk	1140.350 877	2174.96	61.39	0.3508	420.96	35.08	3.508	7.016
Apple	399.88	46.14	1.9994	0.9228	0	35.374	0	0
Millet (bajra)	10799.46	10399.48	205.70 4	131.9934		0	38.2838	0
Ghee	2778.9	12.56	0.942	0	27.632	0	0.3768	0
Butter	2050.62	68.64	1.43	0	0	0	0	0
Vegetable oil	5429.38	0	0	0.0667	0	0	0	0
Oil Seeds (mustard)	8320.58	7536.2	307.6	123.04	0	0	0	0
Almond	609.9939	262.2196	22.222	4.55551	0	0	0	0
Common bean	2233.311	846.6582	153.53 1798	33.333	0	0	0	0
Wheat	5827.12	586.16	241.36	63.9604	0	0	0	0
Rice	2866.38	222.2	59.327 4	26.664	0	0	0	0

Part II - Output

Positive variable $x(f)$: Amount (in rupees) to spend on food (f) per day. The model provides the optimal amount that must be spent on each food item to fulfill the RDA.

Free variable (cost): This is the total amount required to be spent on the optimal diet on a daily basis (the sum of all values of $x(f)$). The model seeks to minimize this variable.

Nutrient balance $nb(n)$ (in units)

Equation: $\sum(f, a(f,n)*x(f)) \text{ =g= } b(n)$;

The equation states the total amount of nutrient 'n' consumed from all foods weighted by how much money is spent on them. This should be greater than or equal to the RDA. This equation ensures that the total nutrients

obtained from all chosen foods are at least as much as the required amount for each nutrient.

(where, “=g=” is a GAMS syntax which means greater than or equal to.)

where $a(f,n)$ = amount of nutrient n contained in one unit of food f. $x(f)$ = decision variable = how much quantity of food f to include. $b(n)$ = minimum required intake of nutrient n.

Cost balance (cb) (in rupees) - Equation : $\text{cost} \text{ =e= } \sum(f, x(f))$;

It ensures that the variable cost correctly reflects the total amount spent on food for a day based on how much money is spent on each food item.

(where, “=e=” is the GAMS syntax for equality.)

It serves as the link between the amount spent on individual food items ($x(f)$) and the total daily cost to be minimised in the objective function.

Application

The table construction of the Nutrient Value of Foods (per ₹100) table, $a(f,n)$, has three separate steps for each food-nutrient pair. The conversion is required because food price serves as the denominator that translates raw amounts of nutrients into economic efficiency indicators.

Step 1: Defining the Constraints ($b(n)$) - The ICMR Mandate

The Required Daily Allowances ($b(n)$) are directly taken from the ICMR-NIN (2020) Expert Group Report. This sets the minimum targets to be achieved.

The ICMR report is the ultimate reference for Indian dietary guidelines. The figures such as 1660 Kcal (Estimated Energy Requirement), 1000 mg Calcium, and 46 g Protein are taken directly, with the aim of ensuring the constraints of the model are clinically relevant to the target population. The model employs 140 μ g for Iodine, 2.2 μ g for Vitamin B 12

Step 2: Collecting Raw Nutrient Data (Nutrient per 100g)

This was done by cross-referencing food composition tables in order to find the raw nutrition prior to cost. Staples and Pulses (Millets, Wheat, Common Bean): Resources such as the ICMR-NIN's Indian Food Composition Tables (IFCT 2017) and technical reports on millets (e.g., ilsa-india.org) were crucial. These local resources give precise data for particular Indian cultivars, revealing, for example, the very high Calcium and Fiber in Bajra (Millets).

Animal Products and Vitamins (Milk, Eggs, Meat): Global data, i.e., the USDA Food Data Central (FDC) and specialized reports (milk.co.uk, egginfo.co.uk), were employed to corroborate or verify Vitamin B

12, Niacin, and Iodine content since they are essential and in short supply in plant-based diets. The Vitamin B 12, within eggs and meat is a particularly significant input.

Micronutrient-Dense Foods (Green Leafy Vegetables, Orange): The rich Vitamin C in Orange and the high Iron and Calcium in Green Leafy Vegetables were established based on typical composition data.

Step 3: The Economic Change ($a(f,n)$)

If Bajra is economical and rich in Calcium, it will provide an enormous value (e.g., 10399.48 mg for ₹100). It means that Millets is a very effective, low-price source for fulfilling the Calcium limitation. Meat contains a high Niacin and Protein content, but at a high cost.

The Strengths of the Model in an Indian Context

1. **Cost Sensitivity:** Reducing cost, the model speaks directly to a base concern in Indian families, particularly in poor households.
2. **Micronutrient Focus on Deficiencies:** The model encompasses the important nutrients that are typically lacking in the Indian diet (e.g., Iron, Vitamin B 12, Iodine). The average adult woman's high RDA of Iron (29 mg) for losses during menstruation is a limiting factor that will determine much of the diet structure.
3. **Local Food Inclusion:** Inclusion of staple and historically inexpensive foods such as Millets (Bajra), Common Bean, and Green Leafy Vegetables enables the optimization to identify locally viable and low-cost solutions. Because Millets and Common Bean yield the most Iron and Protein per ₹100, they are most likely to be prominent in the optimal, lowest-cost diet.

Interpretation and findings

From the Classical Stigler's Diet Problem, which was designed to get the minimum cost that fulfills the recommended daily allowances (RDA) of essential nutrients

The model took into picture 15 different food sources that were milk, orange, apple, meat, spinach, eggs millets, ghee, butter, vegetable oil, almond, oilseeds, wheat, rice and common bean which had a constraint of meeting the minimum nutrients that is 1660 kcal of energy, 46 g of protein, 1000 mg of calcium, 29 mg of iron, 140 µg of iodine, 65 mg of vitamin C, 11 mg of niacin, and 2.2 µg of vitamin B12. with the least amount of cost.

The model shows the optimal cost of Rs 154.05, which satisfies a female's daily requirement. The diet is heavily made up of millets (Rs. 153.67), which supply calories, protein, calcium, and niacin. Minimal contributions come from milk (Rs. 0.33), which provides calcium and vitamin B-12, and spinach (Rs. 0.044), which provides vitamin C and iron. All other food items were excluded as they were not cost-effective.

The model has dropped food variables, mainly meat and eggs, as they cost a lot, but they also provide a good quantity of protein, B12, and niacin. However, as our model is optimizing cost, we take what is the cheapest while providing the required amount of nutrients.

Policy Implications & Output Recommendations

The results obtained in the present study have significant policy implications on nutrition, development of food subsidies, and community health promotion in India. The use of Linear Programming (LP) in GAMS indicates that through the cost-optimal diets, full nutritional adequacy of women could be attained provided locally available and culturally relevant foods especially millets, pulses, and leafy vegetables were given priority. This understanding implies that the national nutrition strategies should be reassessed to be less calorie-sufficient and more nutrient-dense and affordable with Re-focusing food subsidies and public distribution systems (PDS), the Current food security programs in India are still largely based on the procurement of rice and wheat. The model demonstrates that the traditional staples like millets and pulses offer better returns of nutrients such as calcium, protein and iron at significantly higher cost efficiency. The inclusion of these products in the PDS basket and higher procurement incentives to farmers of millet and pulse products can be used to enhance both nutrition and agricultural sustainability. The results of the LP would allow pilots to introduce region-specific low-cost nutritious food baskets that will not only satisfy the energy needs, but also the micronutrient requirements. Institutional nutrition-sensitive meal planning, the findings indicate the possible benefit of optimization models in enhancing the cost-effectiveness of meals offered in mid-day meal programs, Integrated Child Development Services (ICDS) and hospital diets. The LP-based structure allows institutions to redesign the menu so that it includes Recommended Dietary Allowances (RDA) on specific groups of women, children, or patients with minimal daily costs. Periodically updated price and nutrient information may serve as an input into an online GAMS-like tool to assist governmental diet planners. This strengthens the women nutrition programs as the model focuses on a 55-kg moderately active Indian woman therefore the policy implications directly relate to nutritional empowerment of women. Women with low income groups have several constraints to balanced diets, including low costs as well as awareness. Programs like Poshan Abhiyaan and Jan Andolan can be made to lobby using the LP results in favor of millet-based, low-cost meals meeting the requirements of the RDA. These optimal food combinations can be trained to community kitchens, self-help groups, and Anganwadis in

provision of meals in day to day activities. As a contribution to the localized data and technological integration to ensure the policy remains in use, the government and research institutions must create an open database combining Indian Food Composition Tables (IFCT 2017) with the updated regional prices. This kind of platform may be useful in assisting policymakers to make cost and nutrient trade-offs in real time, so it may be possible to alter subsidies or dietary guidelines based on the local food economics. Striking a balance between cost efficiency and diversity and sustainability though the optimal diet in the model was mostly composed of millets, overdependence on a single source of food is not realistic. Thus, the policy formulation program must incorporate combinations of cost efficiency of LP and secondary goals in terms of dietary diversity, taste acceptability, and environmental sustainability. A multi-objective optimization model can optimize both cost and nutrition and cultural feasibility to achieve more realistic results in national dietary guidelines.

Overall, this paper demonstrates that computational nutrition modeling can help ensure that the current food security policies, which are more generalized and rely on calories, can be modified to achieve the desired outcome of more nutrient-centered policies. Such optimization tools, embedded within the nutrition and agricultural policies of India could be used to establish resilient, affordable and health-conscious food systems particularly those that are critical towards enhancing the nutrition of women and the overall aspiration of national food security.

Conclusion

The current research was able to utilize the classical Stiglers Diet Problem through the application of Linear Programming (LP) in the General Algebraic Modeling System (GAMS) to calculate the most cost-effective diet of a 55-kg Indian woman of moderate activity. The model lowered the cost of food intake by incorporating locally sourced food products, nutrient profiles, and price information and by adhering to the Recommended Dietary Allowances (RDA) given by the Indian Council of Medical Research (ICMR-NIN). The best solution was to find a daily diet of 154.05 in the form of millets with a small amount of milk and leafy vegetables. The result underscores the affordability and nutritional sufficiency of the traditional, locally obtained foods to address the everyday needs of nutrients.

Nonetheless, the findings also indicated that only cost minimizing models might produce monotonous culturally implausible diets which are single source only. Thus, future employment ought to build upon the model by adding other goals like dietary variety, palatability and sustainability. Cost, nutritional adequacy and acceptability can be balanced by applying techniques like multi-objective optimization or genetic algorithms. Policy wise, this study shows that data-driven optimization can be used to reinforce national nutrition policies. The subsidies programs must not be based on rice and wheat but on pulses, millets, and vegetables, foods that are nutritious and affordable. Similar LP models could be applied to schools, hospitals and government meal schemes to design low-cost, nutritionally balanced menus. In general, this study is an economics-nutrition science gap filler that demonstrates that optimization tools such as GAMS can offer evidence-based information when planning individual diets and health policies. The utilization of such models can help India to transition to cost-effective, culturally appropriate and health-oriented food systems that can empower women nutrition and serve as a source of national food security.

References

- Indian Council of medical research corporation. (2011). *National Institute of nutrition*. Retrieved from <https://www.nin.res.in/downloads/DietaryGuidelinesforNINwebsite.pdf>
- Milk - THE NUTRITIONAL COMPOSITION OF DAIRY PRODUCTS. (2025, August 29). THE NUTRITIONAL COMPOSITION OF DAIRY PRODUCTS. <https://milk.co.uk/nutritional-composition-of-dairy/milk>
- USDA FoodData Central. (n.d.). <https://fdc.nal.usda.gov/>
- Nutrition Facts for Wheat flour, whole-grain, soft wheat. (n.d.). My Food Data. <https://tools.myfooddata.com/nutrition-facts/168944/100g>
- Australian Food Composition Database. (n.d.). <https://afcd.foodstandards.gov.au/fooddetails.aspx?PFKID=F004205>
- TNAU Agritech Portal :: Post Harvest Technology. (n.d.). <https://share.google/Sm48dZ3dzsKXXt88n>
- Rd, A. S. M. (2024, February 29). Top 12 foods that are high in vitamin B12. Healthline. <https://www.healthline.com/nutrition/vitamin-b12-foods#eggs>
- Vitamins in eggs, minerals in eggs | Official egg info UK. (n.d.). <https://www.egginfo.co.uk/egg-nutrition-and-health/egg-nutrition-information/vitamins-and-minerals>
- U.S. Department of Agriculture, Agricultural Research Service. (2023). *FoodData Central*. FDC ID: 1750342. Retrieved October 26, 2023, from <https://fdc.nal.usda.gov/food-details/1750342/nutrients>
- Samtiya, M., Al-Ghazali, M. A., & Al-Malki, A. L. (2017). Nutritional and Health Benefits of Millets. *International Journal of Molecular Sciences*, 18(2), 255. <https://doi.org/10.3390/ijms18020255>
- T, Longvah., Ananthan, R., Bhaskarachary, K., & Venkaiah, K. (2017). *Indian Food Composition Tables 2017*. National Institute of Nutrition, Indian Council of Medical Research. <https://www.nin.res.in/ebooks/IFCT2017.pdf?utm>

Shobana, S. (n.d.). *Nutritional Composition And Health Benefits Of Millets*. MDRF, Chennai. Retrieved October 26, 2023, from https://ilsi-india.org/Conference_on_Whole_Grains_for_Promoting_Health/Nutritional%20Composition%20And%20Health%20Benefits%20Of%20Millets%2C%20Dr.%20S%20Shobana%2C%20Scientist%2C%20MDRF%2C%20Chennai.pdf?utm_

Eat Right India. (n.d.). *Millets - Nutritious Cereal of India*. Retrieved October 26, 2023, from <https://eatrightindia.gov.in/millets2023/assets/documents/Millets%20-%20Nutritious%20Cereal%20of%20India.pdf>

Appendix

Code for running the gams output-

https://docs.google.com/document/d/1G5FPyX1UOIKK0VWvquJW5Coe6OV8dWrD/edit?usp=share_link&ouid=109152930682059986053&rtpof=true&sd=true